



PROJECT REPORT

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Project Name:

Personal AI Assistant

Submitted To:

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GitHub Repository:

https://github.com/anishaa-mitu/Personal_AI_Assistant

Table of Contents

Abstract.....	2
1. Introduction	3
1.1 Background	3
1.2 Problem Statement	3
1.3 Aim of the Project	3
2. Objectives	3
3. System Overview.....	4
4. System Architecture	5
4.1 Input Layer	5
4.2 Decision Layer	5
4.3 AI Processing Layer	6
4.4 Tool Execution Layer	6
4.5 Output Layer	6
5. Functional Requirements	7
5.1 Gmail Management	7
5.2 Google Calendar Management	8
5.3 Google Drive Management	9
5.4 Web Search	10
5.5 Calculator	12
5.6 Pomodoro Productivity System	12
5.7 Conversational Memory	13
6. Technologies Used	13
7. Implementation Details.....	14
8. Evaluation.....	14
8.1 Strengths.....	14
8.2 Limitations	15
9. Proposal and Final Delivery	15
10. Future Scope.....	15
11. Conclusion.....	16
12. Recommendations	17
13. Technical Notes	18
14. Overall Assessment	18

Abstract

This project presents a Telegram-based personal AI assistant developed using n8n workflow automation and LangChain-based AI agent functionality. The system is designed to interpret natural-language messages and execute actions across multiple services, including Gmail, Google Calendar, Google Drive, web search, and calculation. In addition, it includes a Pomodoro-style productivity workflow that logs study sessions in Google Sheets and sends completion alerts through Telegram. The objective of the system is to reduce digital friction by allowing the user to manage routine tasks from a single conversational interface rather than switching between separate applications. The assistant uses GPT-5 mini as its language model and maintains a 70-message sliding memory window to preserve recent context for each user. The resulting workflow demonstrates how automation, artificial intelligence, and messaging platforms can be integrated into a practical personal productivity assistant.

1. Introduction

1.1 Background

Modern digital environments require users to interact with applications for communication, scheduling, file management, information retrieval, and productivity tracking. Although each of these tools is useful individually, the repeated need to move between them creates inefficiency and interrupts concentration. This problem is especially significant for students and knowledge workers who depend on frequent coordination across emails, calendars, and cloud storage. This project was developed to address that fragmentation by creating one conversational assistant that can manage several digital services through Telegram. Instead of requiring explicit commands or navigation through separate interfaces, the assistant receives a natural language message and decides which operation should be performed.

1.2 Problem Statement

The central problem addressed by the project is the inefficiency caused by app switching and fragmented digital workflows. Routine tasks such as sending an email, checking a calendar, searching for files, or performing a quick calculation often require opening several different platforms. This not only wastes time but also increases cognitive load. A related problem is the lack of a unified natural language interface that can coordinate across productivity services reliably. The project seeks to solve this by combining AI reasoning with workflow automation in a Telegram-based assistant.

1.3 Aim of the Project

The aim of the project is to build a personal AI assistant that can receive a message in Telegram, determine the required action, and execute the appropriate task automatically. The assistant is intended to function as a single point of control for routine productivity and academic tasks.

2. Objectives

The project was developed with the following objectives:

- To design a Telegram-based personal assistant that accepts natural language input.
- To automate common Gmail operations such as sending, replying, drafting, labeling, deleting, and reading emails.

- To automate Google Calendar operations such as event creation, retrieval, and deletion.
- Google Drive operations, such as searching, uploading, moving, and updating files, can be automated .
- To enable real-time web searches using SerpAPI for current information and research-related queries.
- To provide a calculator tool for arithmetic and numerical processing.
- To implement a Pomodoro productivity feature that logs study sessions and sends completion alerts.
- To preserve short-term conversational memory for improved interaction continuity.

3. System Overview

The workflow begins with a Telegram Trigger node that listens for incoming message updates. Once a message is received, the workflow first checks whether the message contains the word “focus.” If the word is present, the message is routed to a Pomodoro branch. Otherwise, the message is sent to the AI assistant branch for normal processing. The AI assistant branch uses a LangChain-powered agent connected to multiple tools. These tools include Gmail, Google Calendar, Google Drive, SerpAPI, and a calculator. The agent is instructed through a system prompt to always perform actions rather than merely describing them. It also follows specific rules for email formatting, calendar scheduling, and tool selection. The Pomodoro branch is separate from the AI path. It starts a 25-minute waiting period, appends a study record to Google Sheets, and sends an alert back to Telegram when the timer ends. This makes the workflow suitable for monitoring academic productivity.

4. System Architecture

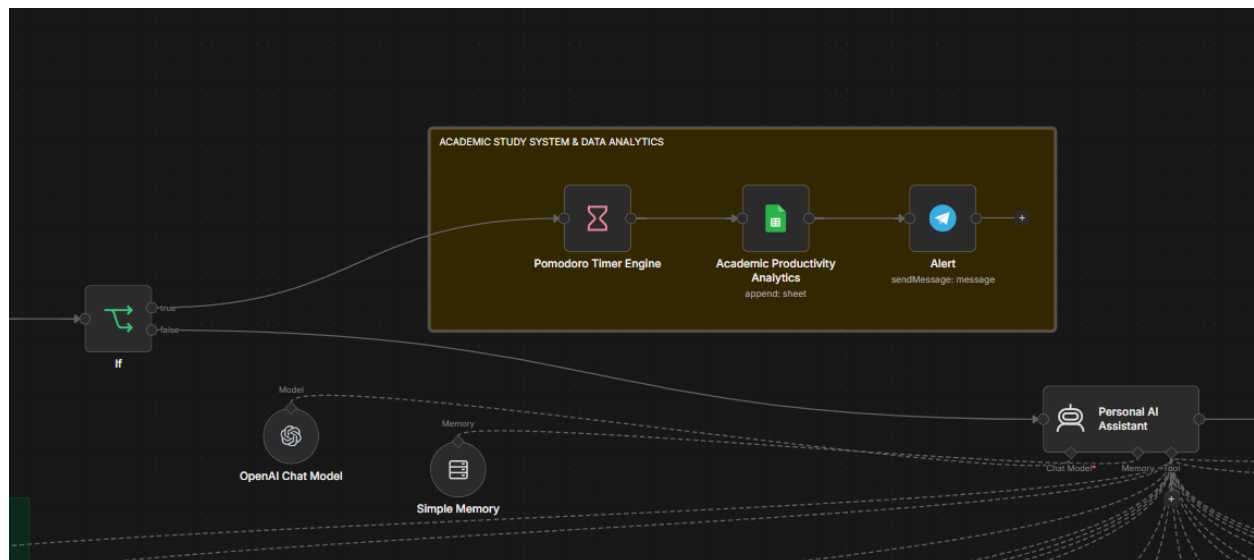
4.1 Input Layer

The system receives messages via the Telegram Bot API. Telegram serves as the only user interface, keeping the interaction simple and accessible. The bot specifically listens for message updates.

4.2 Decision Layer

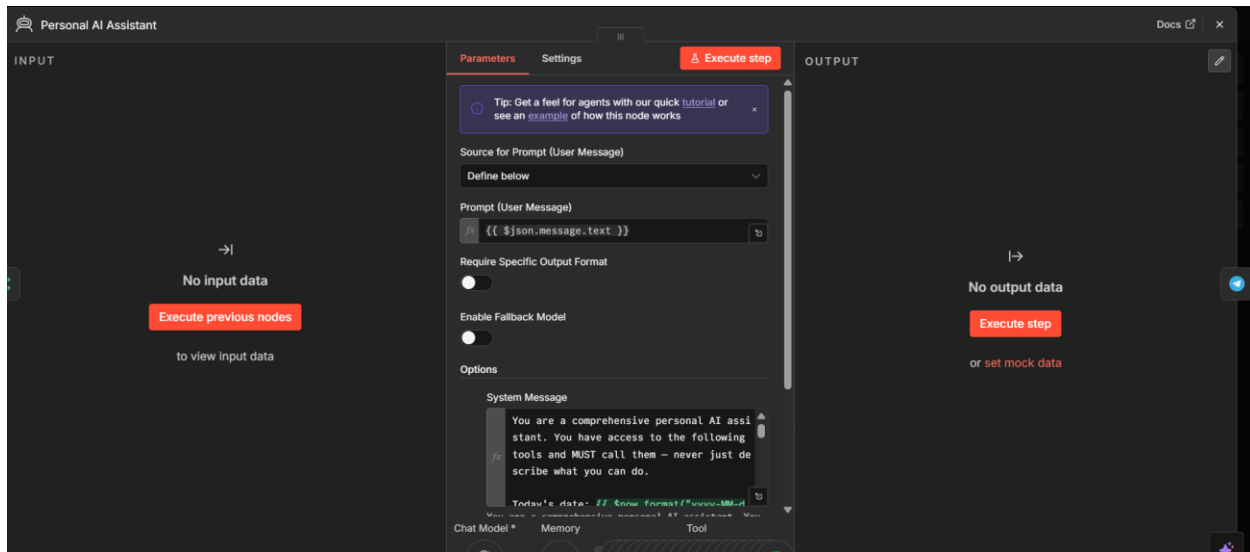
The IF node functions as the first routing mechanism. It examines the message text and checks whether it includes the keyword “focus.” This decision determines whether the message should follow the productivity branch or the AI tool branch.

```
{{ $json.message.text.toLowerCase().includes('focus') }}
```



4.3 AI Processing Layer

If the message does not belong to the focus branch, it is passed to the AI Agent node. This node is powered by GPT-5-mini and configured with a memory buffer window of 70 messages. The agent uses the message context and the system instructions to select and call the correct tool.



4.4 Tool Execution Layer

The assistant is connected to several categories of tools:

- Gmail tools for email management.
- Calendar tools for scheduling tasks.
- Drive tools for file management.
- SerpAPI for web search.
- Calculator for numerical operations.

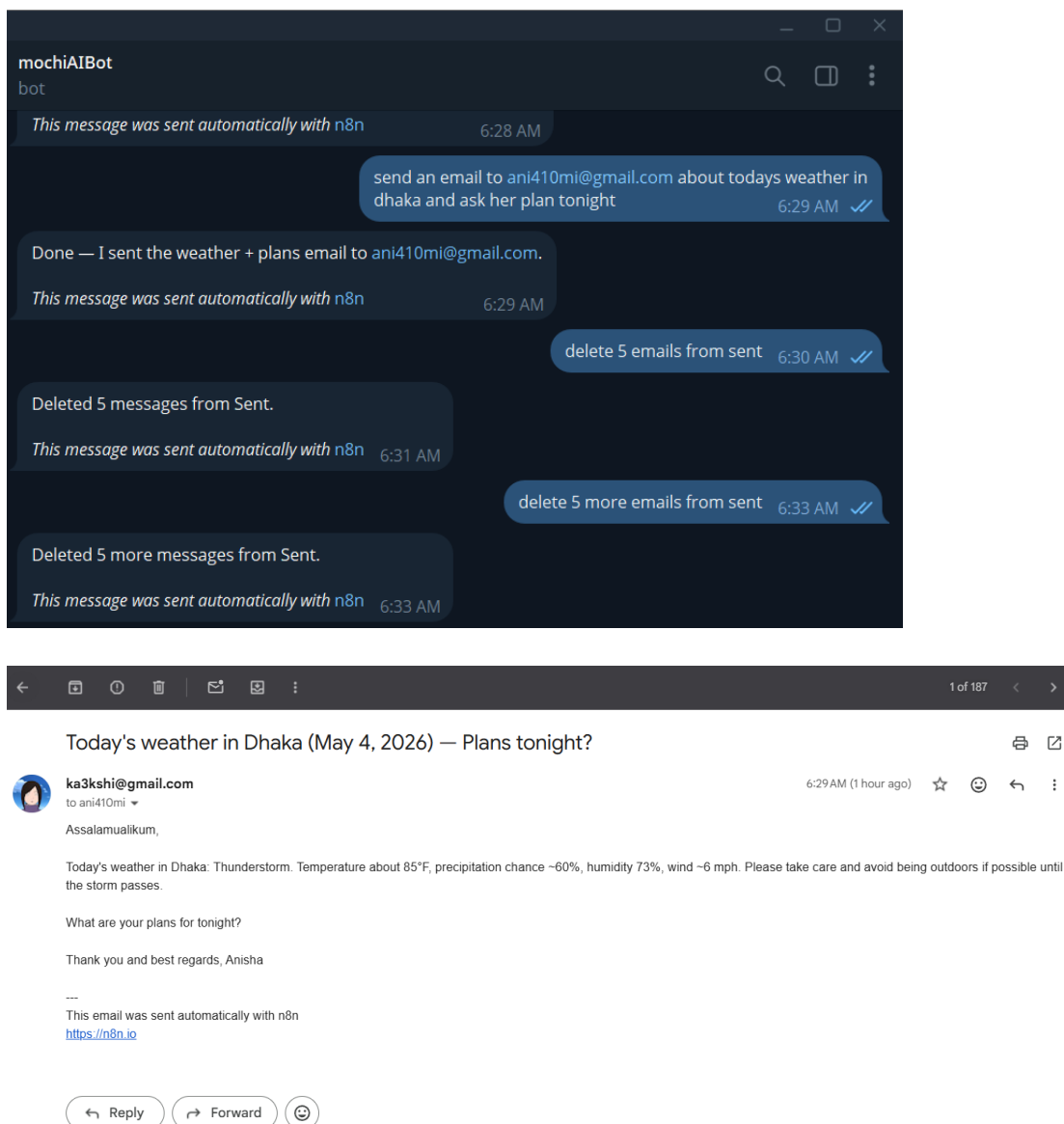
4.5 Output Layer

After the action is executed, the result is returned to Telegram through a text message node. This ensures that the user receives feedback in the same chat where the request was made.

5. Functional Requirements

5.1 Gmail Management

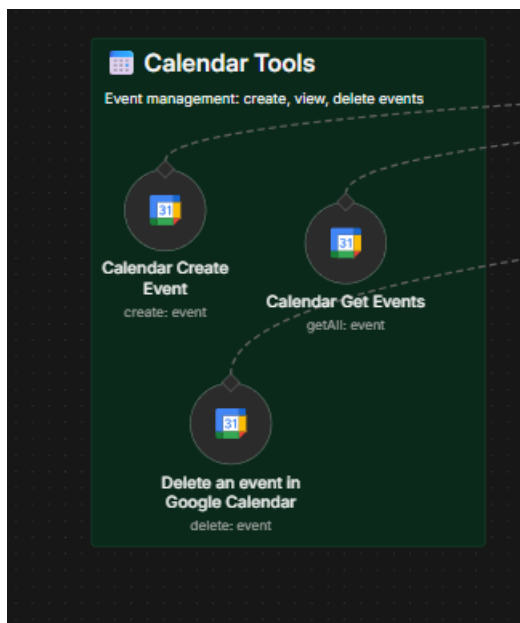
The workflow includes a comprehensive set of Gmail tools. These support sending emails, replying to messages, creating drafts, deleting messages, retrieving inbox items, adding labels, removing labels, marking messages as read or unread, and creating new labels. All outgoing emails are required to begin with “Assalamualikum,” and conclude with “Thank you and best regards, Anisha.” The prompt distinguishes between simple and important emails. Simple messages can be sent directly, whereas important messages should be created as drafts for user review before delivery. This design reflects a practical balance between automation and the user’s control.

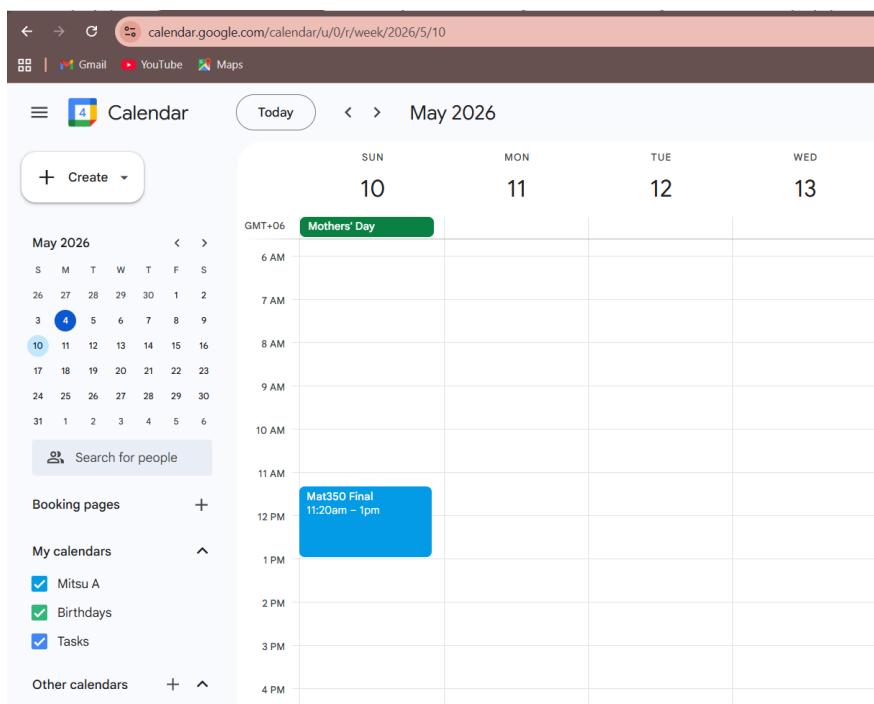
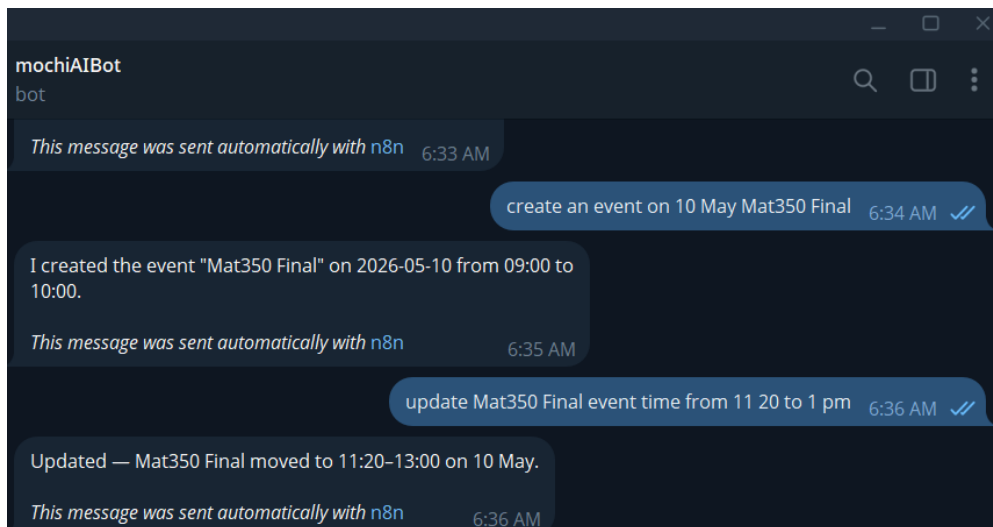




5.2 Google Calendar Management

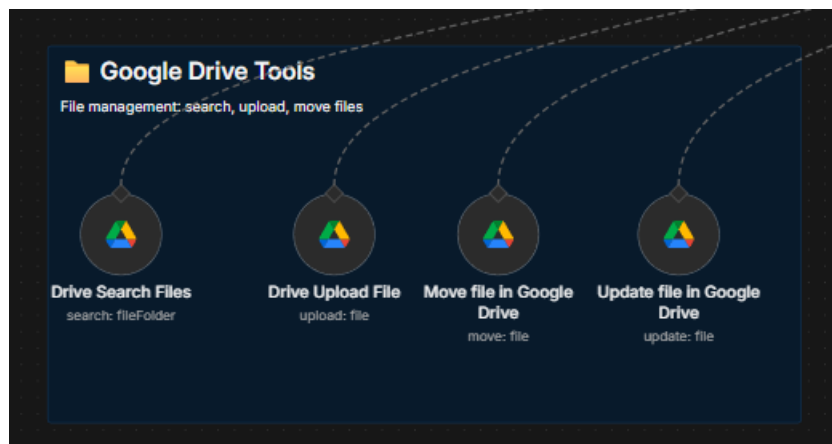
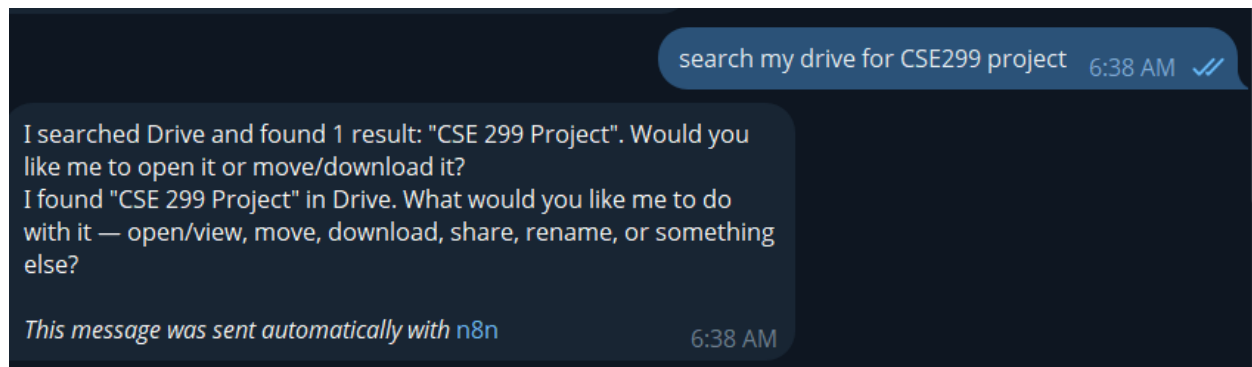
The assistant supports the creation, retrieval, and deletion of calendar events. When a user requests an event, the AI infers the start and end times from the natural language message. If no end time is provided, the system should automatically set the end time to one hour after the start time. The workflow also includes a calendar retrieval node that returns up to 20 upcoming events within the next seven days, ordered by the start time. This makes it possible to quickly check schedule information from Telegram.





5.3 Google Drive Management

The assistant can search for files or folders in My Drive, upload files, move files between folders, and update file content or metadata. This allows the user to perform document related tasks without manually opening Google Drive.



5.4 Web Search

The web search tool uses SerpAPI with desktop and English locale settings. It is intended for real-time queries involving current news, research, nutrition, and general factual information. This tool extends the assistant beyond static knowledge and enables access to current web based content.

SerpApi

APIs

Integrations

Features

Pricing

Use cases

Company

Sign In

Register

APIs

Explore all search engines we support or check out our complete API list.

Google Search API

Google Light Search API

Google AI Mode API

Google AI Overview API

Google Ads Transparency API

Google Autocomplete API

Google Events API

Google Finance API

Google Finance Markets API

Google Flights API

Google Flights Autocomplete API

Google Forums API

Google Hotels API

Google Hotels Autocomplete API

Google Hotels Photos API

Google Hotels Reviews API

Google Images API

Google Images Light API

Google Immersive Product API

Google Jobs API

Google Lens API

Google Local API

Google Local Services API

Google Maps API

Google Maps Autocomplete API

Google Maps Contributor Reviews API

Google Maps Directions API

Google Maps Photos API

Google Maps Posts API

Google Maps Reviews API

Google News API

Google News Light API

Google Patents API

Google Patents Details API

Google Play Books API

Google Play Games API

Google Play Product API

Google Play Movies API

Google Play Product API

Google Play Store API

Google Related Questions API

Google Reverse Image API

Google Scholar API

Google Scholar Author API

Google Scholar Case Law API

Google Shopping API

Google Shopping Light API

Google Short Videos API

Google Travel Explore API

Google Trends API

Google Trends Autocomplete API

Google Trends Trending Now API

Google Videos API

Google Videos Light API

Amazon Search API

Amazon Product API

Apple App Store API

Apple Maps API

Apple App Store Product API

Apple App Store Reviews API

Baidu Search API

Baidu News API

Bing Search API

Bing Copilot API

Bing Images API

Bing Maps API

Bing News API

Bing Product API

Bing Reverse Image API

Bing Shopping API

Bing Videos API

Brave AI Mode API

DuckDuckGo Search API

DuckDuckGo Light API

DuckDuckGo Maps API

DuckDuckGo News API

eBay Search API

eBay Product API

Facebook Profile API

The Home Depot Search API

The Home Depot Product API

The Home Depot Reviews API

Naver Search API

Naver AI Overview API

OpenTable Reviews API

Tripadvisor Search API

Tripadvisor Place API

Tripadvisor Reviews API

Walmart Search API

Walmart Product API

Walmart Reviews API

Yahoo! Search API

Yahoo! Images API

Yahoo! Shopping API

Yahoo! Videos API

Yandex Search API

Yandex Images API

Yandex Videos API

Yelp Search API

Yelp Place API

Yelp Reviews API

YouTube Search API

YouTube Video API

YouTube Video Transcript API

Extra APIs

Status and Error Codes

mochiAIbot

bot

make me a diet plan for 85 year old prostate cancer patient

06:48 AM ✓

I looked up current nutrition guidance for older adults with cancer and made a safe, practical plan you can use or personalize. Before I calculate exact calories/protein I need the patient's weight, height, current treatments (e.g., hormone therapy, chemo, radiation), appetite issues, swallowing problems, allergies/intolerances, and goal (maintain weight / gain / lose). Please provide those if you want a tailored plan. Meanwhile, here's a general, evidence-based plan and key recommendations:

Key targets and principles

- Energy: aim ~25–35 kcal/kg/day depending on activity and nutrition status (use the lower end if sedentary, higher if malnourished or losing weight).

- Protein: 1.0–1.5 g/kg/day (many guidelines recommend ~1.2 g/kg/day for older cancer patients; increase toward 1.5 g/kg if weight loss or muscle loss).

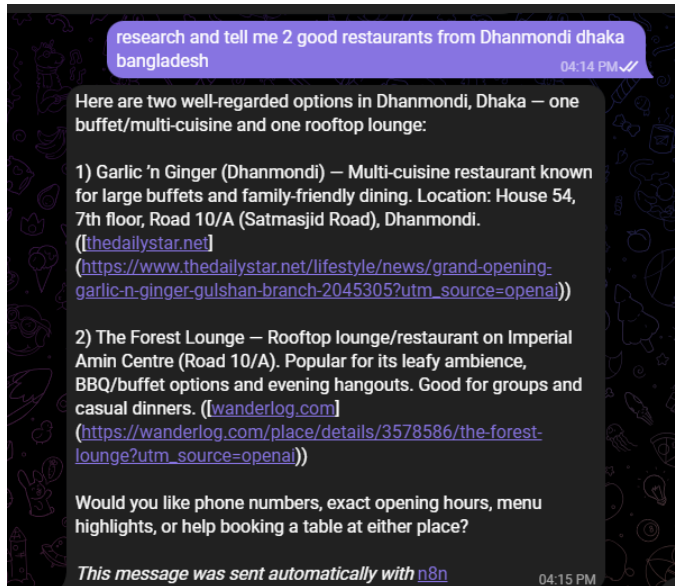
- Pattern: small frequent meals (3 meals + 2–3 snacks) to overcome poor appetite.

- Focus on nutrient-dense, easy-to-eat foods: soft proteins (eggs, yogurt, tofu, soft fish), smoothies, fortified soups.

- Follow a Mediterranean-style pattern where possible: lots of vegetables, whole grains, legumes, healthy fats (olive oil, nuts), limited processed/red meat and refined sugars.

- Food safety/hygiene: avoid raw/undercooked animal products.

Message

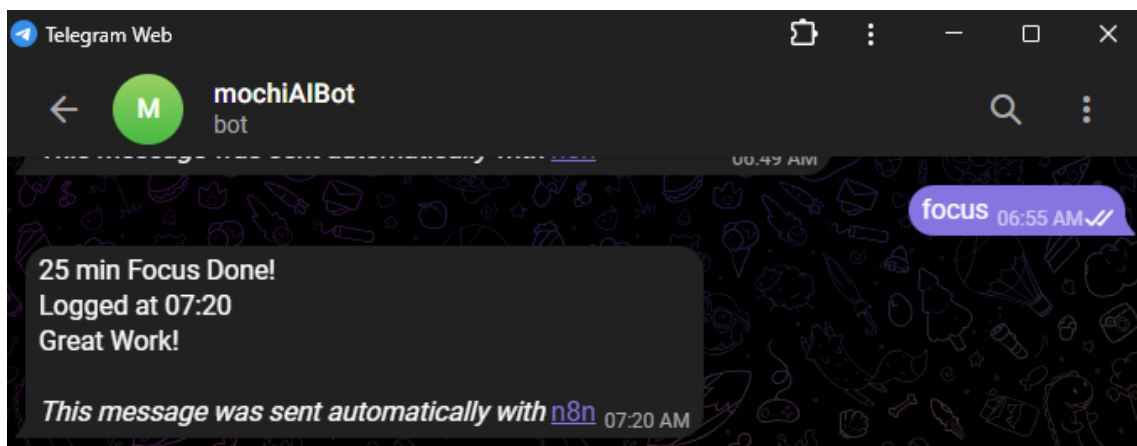


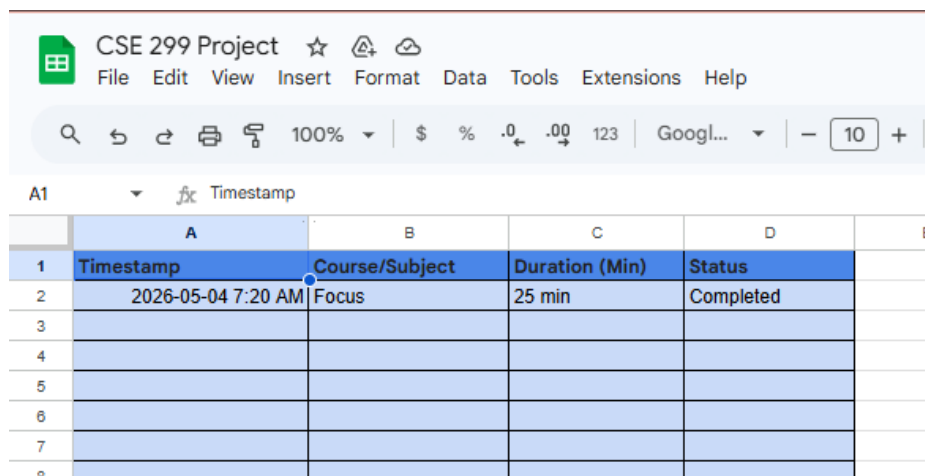
5.5 Calculator

The calculator tool handles arithmetic and numerical tasks. It is intended for use in situations where the assistant must compute calorie totals, perform conversions, or carry out standard calculations. This improves the reliability of the numerical responses.

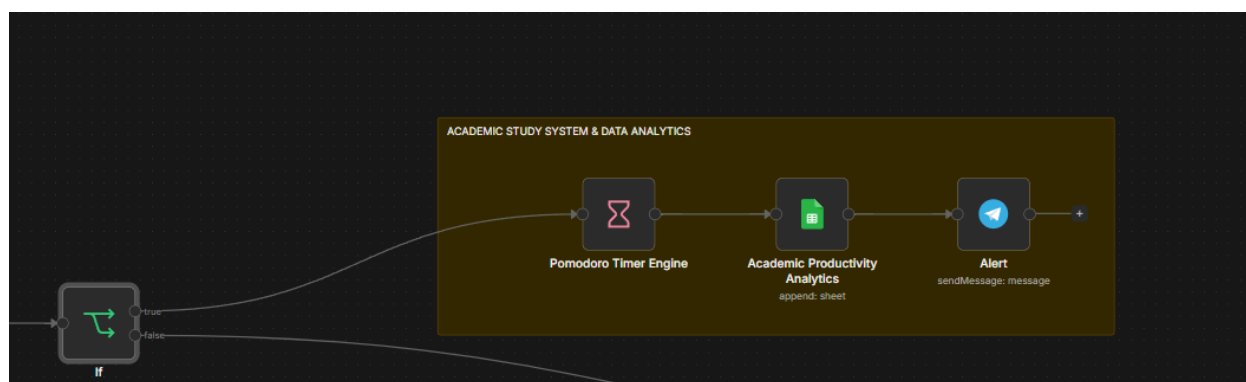
5.6 Pomodoro Productivity System

The focus system is triggered when the user includes the word “focus” in the message. The workflow waits for exactly 25 minutes, logs the session in Google Sheets, and then sends a completion alert through Telegram. The sheet stores timestamp, subject, duration, and status.





	A	B	C	D	E
1	Timestamp	Course/Subject	Duration (Min)	Status	
2	2026-05-04 7:20 AM	Focus	25 min	Completed	
3					
4					
5					
6					
7					
8					



5.7 Conversational Memory

The memory node stores the last 70 messages for each Telegram user. This supports continuity in conversation and improves the usefulness of follow up interactions. However, the memory is session based and does not persist if the workflow or n8n instance restarts.

6. Technologies Used

Component	Technology
Workflow automation	n8n
AI agent framework	LangChain via n8n
Language model	OpenAI GPT-5-mini
Messaging interface	Telegram Bot API

Email service	Gmail API with OAuth2
Calendar service	Google Calendar API with OAuth2
File storage	Google Drive API with OAuth2
Analytics logging	Google Sheets API with OAuth2
Web search	SerpAPI
Memory	n8n buffer window memory

7. Implementation Details

The workflow is structured so that each function is implemented as an independent tool node. This modularity is one of the strongest aspects of the project because it allows the assistant to expand without major structural redesign. New services can be added by attaching more tool nodes to the same AI agent. The system prompt plays an important role in shaping the assistant's behaviour. It gives explicit instructions about how to manage email, calendar, Drive, search, and calculations. It also enforces task-oriented responses and discourages unnecessary explanation. Such instruction design helps the model behave more like an operational assistant than a conversational chatbot. The Pomodoro workflow is separated from the AI agent to ensure reliable execution of timed study sessions. This decision simplifies the control flow and prevents the agent from having to manage the time delay itself. The workflow then records the session in Sheets and informs the user when the focus period ends.

8. Evaluation

8.1 Strengths

The system has several notable strengths. First, it provides a single conversational interface for multiple services, which reduces app switching and supports efficiency. Second, it is modular and easy to extend. Third, it combines productivity automation with academic logging, which gives the project a practical student-oriented purpose. The use of a memory buffer also improves the user experience by preserving recent context. In addition, the use of GPT-5-mini within n8n allows the assistant to reason over tool selection and respond flexibly to a broad range of user requests.

8.2 Limitations

The workflow exported in the file is currently inactive. This means it is not running in production mode and must be activated manually before use. Another limitation is the absence of dedicated error handling. If a tool or API fails, the workflow may terminate without notifying the user.

The system is also bound to one Google account, which means it is not truly multiuser at the credential level. Furthermore, the Pomodoro logging node contains a mapping error because the subject field is currently set to the Telegram chat ID rather than the actual study subject. The assistant also does not support media messages such as images or voice notes.

9. Proposal and Final Delivery

The final build goes beyond the original proposal in several ways. It includes Gmail inbox management, Google Drive tools, real-time web search, calculator support, Pomodoro automation, and sliding-window memory. These additions make the system more capable than a simple email or calendar bot.

Some proposed features were not implemented in the final version. Canvas LMS integration and the dedicated Diet Planner were removed from the final scope. Based on the project documentation, this was due to platform constraints and limited time or resources.

10. Future Scope

The future direction of the project, aligned with the slide content, includes the following enhancements:

1. **AI CV and Job Assistant**

The assistant could be extended to support CV creation, cover letter drafting, job searching, and application tracking. This would make the system useful for career preparation and employment support.

2. **Scam Website Detection**

A security module could be added to detect suspicious links and malicious websites

before the user opens them. This would improve safety in email and messaging environments.

3. **YouTube Video Summarisation**

The assistant could process YouTube links, transcribe the content, and generate concise summaries. This would be useful for lectures, tutorials, and academic research.

4. **AI UI/UX Generation**

The assistant could generate interface concepts, mockups, or visual layout suggestions from text prompts. This would support coursework and project design activities.

5. **Voice Command Support**

Telegram voice notes could be transcribed and used as input for the assistant. This would allow more natural, hands-free interaction.

6. **Multi-user System**

The assistant could be redesigned to support multiple users independently. This would require improved identity handling and dynamic credential management.

7. **Productivity Dashboard**

The study logs stored in Google Sheets could be visualized in a dashboard showing hours studied, streaks, and subject trends. This would convert passive records into actionable insights.

8. **Notion and Trello Integration**

The assistant could be linked with task management platforms such as Notion and Trello. This would allow conversations to create or update tasks directly.

9. **Advanced Error Recovery**

A proper recovery layer could be introduced to catch failures, retry operations, and notify the user clearly when something goes wrong. This would significantly improve reliability and trust.

11. **Conclusion**

This project demonstrates the practical value of combining workflow automation, artificial intelligence, and messaging-based interaction into a single personal assistant. By using Telegram

as the user interface and n8n as the orchestration layer, the system successfully unifies several core productivity functions, including email management, calendar scheduling, file operations, web search, numerical calculation, and academic focus tracking. The result is a conversational assistant that reduces the need for repeated app switching and supports a more efficient digital workflow. The project is also significant because it shows how an AI agent can be configured to act as an operational layer rather than only a conversational interface. The agent does not merely answer questions; it selects tools, executes tasks, and returns results in a compact and practical form. This makes the system especially useful in a student environment, where speed, simplicity, and task consolidation are highly valuable. Although the current implementation has several limitations, including the absence of error handling, the single-account structure, and the temporary nature of memory, the overall architecture remains strong and expandable. The workflow is modular, the interaction model is natural, and the feature set is broad enough to serve as a credible prototype for a more advanced personal AI assistant in future development.

12. Recommendations

Based on the current implementation, the following recommendations are proposed for future improvement:

- Introduce a proper error handling workflow so that failures in external services are detected and reported to the user rather than causing silent termination.
- Replace the current in-memory conversational buffer with persistent storage to preserve context across sessions and restarts.
- Correct the Pomodoro logging field so that the actual study subject is recorded instead of the Telegram chat ID.
- Add support for non-text Telegram inputs such as images, documents, and voice notes.
- Improve the calendar deletion workflow by first retrieving events and allowing deletion by event title or context, not only by event ID.
- Refactor the Google service layer if multi-user deployment is desired in the future.
- Add a productivity dashboard to visualize focus-session data stored in Google Sheets.
- Implement voice-based interaction to make the assistant more flexible and accessible.

These recommendations would improve both the reliability and the academic value of the system. They would also move the project from a functional prototype toward a more mature productivity assistant capable of supporting real world use.

13. Technical Notes

The workflow is currently built around a single Telegram-based entry point. This design keeps the system simple, but it also means that all interactions depend on the messaging platform. In practical terms, the assistant behaves like a command-and-execution system embedded inside a chat interface, which is one of the reasons it feels lightweight and convenient. The AI prompt is highly directive, which is appropriate for an automation-focused assistant. It contains explicit rules for email formatting, calendar behaviour, Drive operations, search use, and calculator use. This prompt design reduces ambiguity and helps the assistant remain action-oriented. However, prompt-based rules alone cannot guarantee consistency, which is why validation and recovery layers would be beneficial in a later version. The system also reflects an important design choice in separating the Pomodoro branch from the AI agent branch. This ensures that time-based focus sessions are handled deterministically, without relying on the model to manage timing logic. From an engineering perspective, this is a sensible division because it improves reliability and maintains an easy to maintain workflow.

14. Overall Assessment

From an academic perspective, this project is a strong example of applied AI workflow integration. This demonstrates an understanding of automation, modular design, tool-based reasoning, conversational memory, and productivity-oriented system design. The combination of these elements creates a realistic and useful assistant, rather than a purely theoretical prototype. The project also has a clear pedagogical value. It shows how different APIs and services can be orchestrated into a single working system and highlights the strengths and weaknesses of current AI-assisted automation. In particular, it reveals that although language models can support intelligent routing and interaction, production quality systems still require proper validation, fault tolerance, and persistence mechanisms.